



SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)



SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: CSA1402

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS:

Total Marks:50

Annexure A

SCENARIO BASED TASK

Code of Conduct:

I Hari Vishnu N / 192524319 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

SCENARIO-BASED TASK

QUESTIONS:

- 1.Scenario: Parsing Table Conflict in LL(1) While designing an LL(1) parser, a student constructs a parsing table and finds multiple entries in a single cell for a non-terminal and input symbol combination. This leads to confusion during parsing. Analyze how the syntax analyzer uses FIRST and FOLLOW sets in this process, identify the reason for the conflict, and explain how it affects deterministic parsing. Conclude by suggesting what decision should be taken to modify the grammar so that it becomes suitable for LL(1) parsing.
- 2.Scenario: Incorrect Parse Tree Generation A syntax analyzer generates a parse tree for the expression $id + id - id$, but the evaluation gives incorrect results due to improper grouping of operators. Analyze how the syntax analyzer constructs the parse tree, identify the issue related to associativity, and explain how grammar rules influence the structure of the tree. Finally, discuss what decision should be taken to enforce correct associativity and ensure accurate parsing.
- 3.Scenario: Invalid Expression Handling A syntax analyzer in a compiler receives the token sequence corresponding to the expression $a + * b$ from the lexical analyzer. Although all tokens are valid individually, the parser fails to generate a parse tree and reports an error. In this situation, analyze how the syntax analyzer identifies the error based on grammar rules, determine the key issue in the token arrangement, and explain how parsing techniques such as predictive or shift-reduce parsing handle such cases. Further, discuss what decision should be taken to improve error detection and recovery while maintaining clarity in parsing.
4. Scenario: Ambiguity in Conditional Statements A syntax analyzer is designed using the grammar $S \rightarrow \text{if } E \text{ then } S \mid \text{if } E \text{ then } S \text{ else } S \mid a$. While constructing the parsing table, conflicts arise, and the parser cannot decide which production to apply for certain inputs. In this scenario, explain how the syntax analyzer encounters ambiguity, identify the key issue known as the dangling else problem, and discuss how it impacts parsing. Suggest an appropriate modification or parsing strategy to resolve the ambiguity and ensure correct association of else clauses.
- 5.Scenario: Operator Precedence Conflict A parser is designed for arithmetic expressions using the grammar $E \rightarrow E + E \mid E * E \mid id$. While parsing the input $id + id * id$, the syntax analyzer produces an incorrect parse tree due to ambiguity in operator precedence. Examine how the syntax analyzer processes this input, identify the issue caused by the grammar, and explain how precedence and associativity rules should be applied. Based on this, suggest what changes must be made to the grammar or parsing technique to ensure correct evaluation.

1. Parsing Table conflict in LL(1):

Answer:-

* In an LL(1) parser, the syntax analyzer constructs the parsing table using the FIRST and FOLLOW sets.

- FIRST set determines the first terminal that can appear from a production.
- FOLLOW set determines the terminals that can appear immediately after a non terminal.

A conflict occurs when than one production is entered in the same parsing table cell. This usually happens because:

- * The grammar is left recursive or
- * Two production have common prefixes, making their FIRST sets overlap.

Due to this conflict, the parser cannot uniquely decide which production to use. Hence, the grammar is not deterministic and is not LL(1).

Decision:- The grammar should be modified by eliminating left recursion and performing left factoring so that each parsing table entry contains only one production, making the grammar suitable for LL(1) parsing.

2 Incorrect parse tree generation :-

Answer :-

* The syntax analyzer constructs a parse tree according to the grammar rules.

* For expression $Pd + Pd - Pd$, an incorrect parse tree may be generated if the grammar does not enforce left associativity. The expression may be grouped as:

- $Pd + (Pd - Pd)$

Instead of;

- $(Pd + Pd) - Pd$;

This production produces incorrect evaluation.

* The grammar determines how the parse tree is formed. If associativity is not specified, different parse trees can be generated.

Decision: The grammar should be rewritten to enforce left associativity, on precedence and associativity rules should be added in the parser to ensure correct parsing and evaluation.

3 Invalid expression Handling :-

Answer:

- * The syntax analyzer reviews the token sequence; $a + * b$
- * Although each token is valid, the sequence does not follow grammar rule.
- * After the operator $+$, the parser expects an operand, but another operator is found $*$. Therefore, no valid parse tree can be generated.
- * In predictive parsing, the parser reports an error because no production matches the input.
- * In shift-reduce parsing, no valid shift or reduce action is possible, resulting in a syntax error.

Decision: The parser should implement error recovery techniques such as panic mode recovery or phrase level recovery to report meaningful errors and continue parsing the remaining input.

4. Ambiguity in conditional statements:-

Answer:-

* Given the grammar

- $S \rightarrow \text{if } E \text{ then } S$
- $S \rightarrow \text{if } E \text{ then } S \text{ else } S$
- $S \rightarrow a$

* The Syntan analyzer faces ambiguity because it cannot determine which production to apply when it encounters else.

* This ambiguity problem is known as the Dangling Else problem.

* The ambiguity cause conflicts during parsing because the parser cannot determine whether the else belongs to the nearest if or an earlier if.

Decision: The grammar should be modified to distinguish matched and unmatched statements, or the parser should follow the rule that else is associated with the nearest unmatched if. This remove ambiguity and ensures correct parsing.

5. Operator precedence conflict :-

Answer:

* Given the grammar:

- $E \rightarrow E + E$
- $E \rightarrow E * E$
- $E \rightarrow Id$

* When parsing the input;

- $Id + Id * Id$

* The grammar is ambiguous because it does not specify operator precedence.

* The parser may generate either:

$$\Rightarrow (Id + Id) * Id$$

(or)

$$\Rightarrow Id + (Id * Id)$$

* leading to incorrect evaluation.

* Since, Multiplication has higher precedence than addition, the correct parse tree is;

$$\Rightarrow Id + (Id * Id).$$

* Associativity should be defined so that operator with the same precedence are evaluated correctly.

Decision: Rewrite the grammar as;

- $E \rightarrow E + T \mid T$
- $T \rightarrow T * F \mid F$
- $F \rightarrow Id$

OR define precedence and associativity rule.



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RUBRICS FOR CALCULATION:

Criteria	Weight age (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand